

Hongcheng Jiang

Applied Scientist | Multimodal AI & Computer Vision

U.S. Permanent Resident | Citizenship Expected Nov. 2026

Open to Remote | Relocation Nationwide | (+1) 626-206-2285

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Professional Summary

Applied AI researcher with a PhD in Electrical and Computer Engineering and industry ML engineering experience. Develops transformer- and diffusion-based models for medical imaging and cross-spectral reconstruction, combining architecture design and controlled evaluation with experience building LLM agents and retrieval systems.

Education

Ph.D., Electrical & Computer Engineering (GPA: 4.0/4.0)

University of Missouri–Kansas City

Jan 2020 – May 2025

Kansas City, MO

M.S., Computer Science (GPA: 3.8/4.0)

University of Missouri–Kansas City

Aug 2017 – Dec 2019

Kansas City, MO

Technical Skills

Modeling: PyTorch, Hugging Face, Transformers, Diffusion Models, Attention, Multimodal Learning

Vision: Medical Imaging, Image Restoration, Super-Resolution, Cross-Spectral Translation, Remote Sensing

Research Evaluation: Baseline Comparisons, Ablation Studies, Patient-Disjoint Splits, Reconstruction Metrics

LLM Systems: LangGraph, Hybrid Retrieval, Reranking, Schema Grounding, Agent Evaluation

Engineering: Python, SQL, NumPy, Pandas, FastAPI, Docker, Git, Linux, vLLM, Ollama

Research & Professional Experience

Research Associate

University of Missouri–Kansas City

June 2026 – Present

Kansas City, MO

- **Medical AI:** Developed GeoVIRA for automated foot and spine angle measurement from radiographs, combining predicted anatomical landmarks with frozen medical image features.
- **Model Engineering:** Designed a relational geometry encoder capturing landmark distances, orientations, and angular relationships; implemented **geometry-conditioned cross-attention** and measurement-aligned visual feature sampling.
- **Data Pipelines:** Prepared radiographic datasets and built training pipelines with patient-disjoint foot splits and **out-of-fold landmark predictions** to avoid in-sample detector outputs during downstream model training.
- **Model Evaluation:** Evaluated geometry-only and geometry-visual models through baseline comparisons, component ablations, and matched-versus-shuffled visual controls to assess the contribution of case-specific image features.

Machine Learning Engineer

NextTier – IT Solutions Consultancy

June 2025 – May 2026

Sacramento, CA (Remote)

- **Applied NLP:** Built a natural-language SQL assistant for revenue analysis, order lookup, and grouped reporting, combining retrieval, agent orchestration, and evaluation.
- **Retrieval & Reasoning:** Implemented **BM25/vector retrieval with cross-encoder reranking** and **LangGraph Planner–Verifier workflows**, with scoped context, independent result checking, clarification, and bounded repair.
- **ML Systems:** Built FastAPI/MCP services with durable asynchronous jobs, worker recovery, read-only query controls, and human-approved mutations backed by transactional idempotency.

Graduate Research Assistant

University of Missouri–Kansas City

Jan 2020 – May 2025

Kansas City, MO

- **Vision & Generative Modeling:** Developed **transformer- and diffusion-based models** for thermal super-resolution, hyperspectral restoration, and NIR-to-RGB translation, evaluating spatial detail and spectral fidelity.
- **Clinical NLP:** Co-developed an **LSTM–CRF** model for medical concept extraction from clinical text, modeling contextual dependencies and structured label sequences.
- **ML Engineering:** Built training and evaluation pipelines covering preprocessing, augmentation, hyperparameter experiments, baseline comparisons, and ablation studies.
- **Research Publications:** Authored and co-authored peer-reviewed papers on image reconstruction, cross-spectral translation, remote sensing, and clinical NLP.

Selected Publications (8 First-Author)

- H. Jiang et al., “GeoVIRA: Geometry–Visual Interfaces for Radiographic Angle Measurement,” *IEEE/CVF WACV*, 2027 (Recommended to 2nd round).
- H. Jiang et al., “SMLP-KAN: Spectral MLP-KAN Diffusion Prior for Hyperspectral Image Restoration,” *IEEE/CVF CVPR Workshop (PBVS)*, 2026.
- H. Jiang and Z. Chen, “Transformer-based Diffusion and Spectral Priors Model for Hyperspectral Pansharpening,” *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 18962–18977, 2025.
- H. Jiang and Z. Chen, “Spatial-Frequency Guided Pixel Transformer for NIR-to-RGB Translation,” *Infrared Phys. Technol.*, vol. 126, Art. no. 105891, 2025.
- H. Jiang and Z. Chen, “Flexible Window-based Self-attention Transformer in Thermal Image Super-Resolution,” *IEEE/CVF CVPR Workshop (PBVS)*, pp. 3076–3085, 2024.

Selected Research & Applied AI Projects

SMLP-KAN — Diffusion-Based Hyperspectral Image Restoration

[GitHub](#)

Python · PyTorch · Diffusion Models · Kolmogorov–Arnold Networks · Attention · Remote Sensing

- Developed an **unsupervised diffusion-based framework** for hyperspectral sharpening, denoising, and inpainting, combining spectral diffusion priors with panchromatic guidance without requiring paired clean-image training data.
- Designed a hybrid **MLP–KAN architecture** with attention-based feature refinement to model spectral–spatial relationships, suppress redundant features, and preserve spatial detail.
- Evaluated restoration across five hyperspectral datasets with component ablations; used **97.6% fewer parameters** than PLRDiff while improving **Botswana** denoising PSNR by **2.21 dB** and SSIM by **4.2%** at $\sigma = 0.1$.

SF-GPT — NIR-to-RGB Image Translation

[GitHub](#)

Python · PyTorch · Transformers · Cross-Spectral Learning · Image Translation

- Developed a **spatial-frequency guided transformer** combining pixel-domain and frequency representations to reconstruct visible-color imagery from near-infrared inputs.
- Designed Spatial-Frequency Guided Multi-head Self-Attention (**SFG-MSA**) to fuse DCT- or DWT-derived features with pixel representations, with channel and spatial attention for texture and structural refinement.
- Evaluated SF-GPT on the **IEEE VCIP 2020 NIR Image Colorization Challenge** dataset; outperformed the **first-place method**, ATcycleGAN, with **1.53 dB higher PSNR**, **2.6% higher SSIM**, and **4.1% lower angular error**.

FW-SAT — Guided Thermal Image Super-Resolution

[GitHub](#)

Python · PyTorch · Transformers · Window-Based Attention · Multimodal Image Fusion · Super-Resolution

- Developed a **transformer-based thermal super-resolution model** combining low-resolution thermal images with aligned high-resolution visible imagery to reconstruct thermal images with finer structural details.
- Designed Flexible Window-based Self-Attention (**FWA**) to aggregate information across neighboring windows, integrating channel and spatial attention with window-based self-attention to capture global, regional, and local features.
- Evaluated thermal super-resolution on the **PBVS 2024 GTISR Track 2** validation set, achieving **4.6%** and **6.9%** higher SSIM than HAN at $8\times$ and $16\times$ upscaling, respectively, with attention ablations.

RadMeasure — Medical Measurement Agent

[GitHub](#)

Python · LLM Agents · Medical Imaging · Geometry Verification · Human-in-the-Loop

- Built a medical measurement agent with a **KEEP/REPAIR/STOP controller** that constrains LLM-proposed actions to registered measurement protocols and tools.
- Implemented geometry verification, bounded repair, human review for unresolved cases, and execution tracing for controlled and auditable measurement workflows.
- Evaluated a selective repair policy offline on 176 cases from **HVAngleEst**, applying corrections to **20.1% of prediction records** and reducing overall MAE from **2.68° to 2.54°** versus no repair.

SQL-Agent — Text-to-SQL Agent & Evaluation

[GitHub](#)

Python · LangGraph · FastAPI · MCP · SQLite/PostgreSQL · Hybrid RAG · Docker

- Built a **LangGraph Planner–Verifier SQL agent** with schema grounding, bounded repair, controlled database execution, and durable FastAPI/MCP services; added Docker deployment and GitHub Actions regression checks.
- Implemented optional **BM25/vector hybrid retrieval and cross-encoder reranking** over schema and business-rule documents, with database/table filtering and separate retrieval evaluation.
- Evaluated on 500 official **BIRD Mini-Dev** questions using full schema context, a Qwen3-4B Planner, and a Qwen2.5-Coder-7B Verifier; achieved **39.2% execution accuracy versus 36.0%** for the local PV-SQL configuration, with **64.5% fewer tokens per correct task** and **3.66 s p95 latency**.